Nonhomogeneous Equations & Undetermined Coefficients

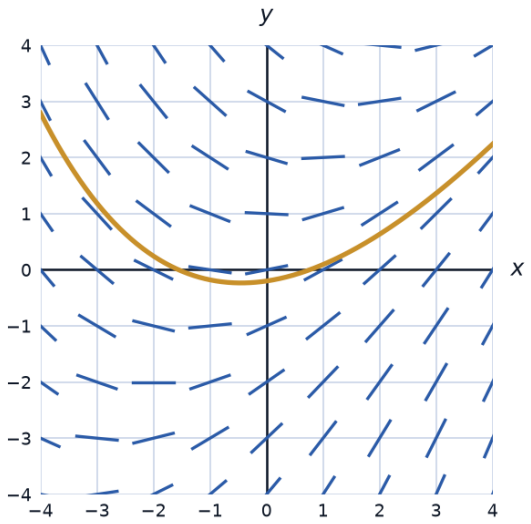
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: General Solution Structure

Find the general real solution of $\mathbf{x}' = (-2, -5; 1, 0)\mathbf{x}$, classify the origin, and determine whether trajectories rotate clockwise or counterclockwise.

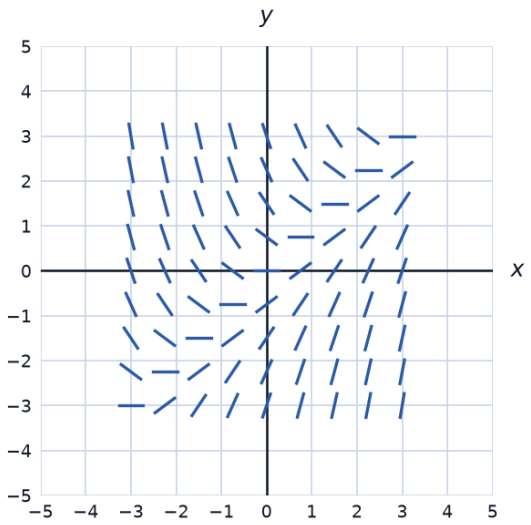
2 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

Consider the differential equation $y'' - 3y' + 2y = f(t)$ with initial conditions $y(0) = 0$ and $y'(0) = 0$. If the forcing function is $f(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, what is the Laplace transform of the solution $Y(s) = \mathcal{L}\{y(t)\}$?



3 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

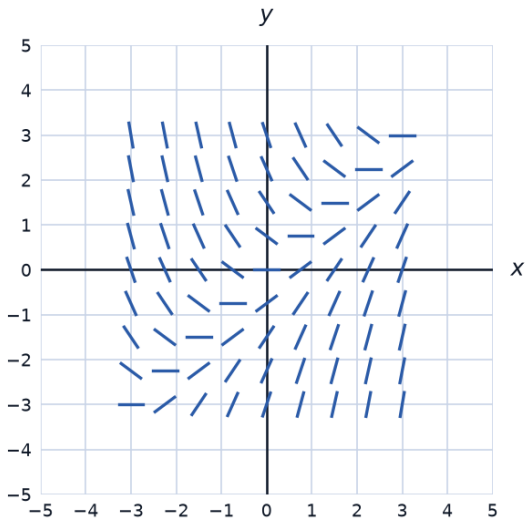
Solve the double-resonance equation $y'' - 6y' + 9y = 4e^{3t}$ with $y(0) = 1$ and $y'(0) = 3$.



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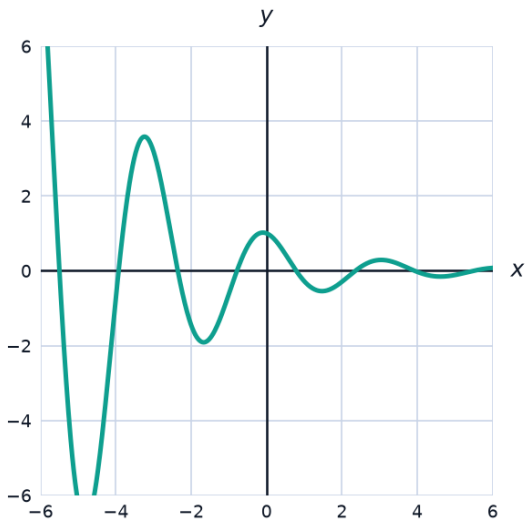
4 [Foundational] • Target: The Principle of Superposition

Construct the Green's kernel $G(x, t)$ for $L[y] = y'' - y$, and express the particular solution of $y'' - y = \frac{1}{e^x + 1}$ satisfying $y(0) = 0, y'(0) = 0$ as an explicit definite integral.



5 [Foundational] • Target: General Solution Structure

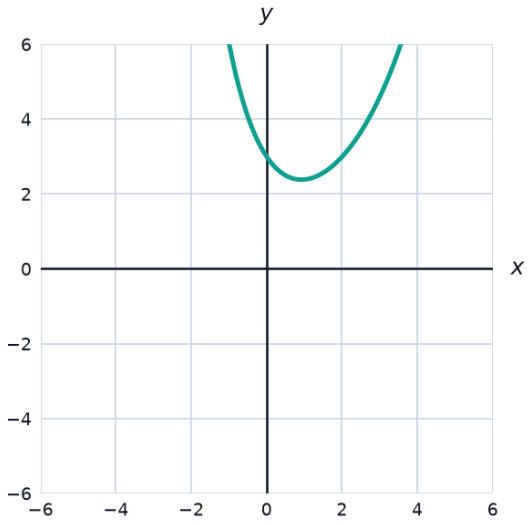
Find a particular solution of $y'' + 4y = \sec(2x)$ for $x \in (-\pi/4, \pi/4)$ using Variation of Parameters.



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6 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

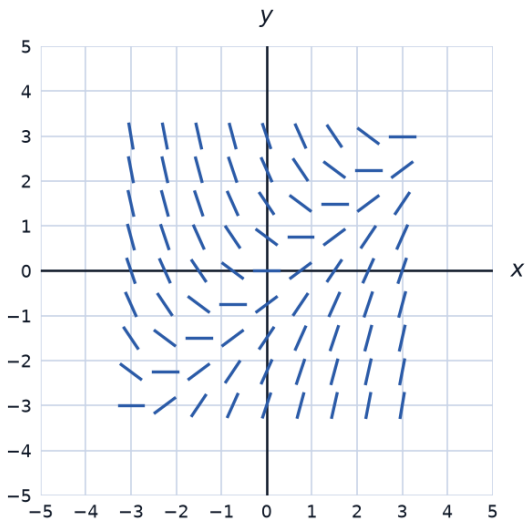
Find a particular solution of $y'' + 2y' + y = \frac{e^{-x}}{x}$ for $x > 0$ using Variation of Parameters.



7 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

Solve the initial value problem with step forcing using the Laplace transform:

$y'' + 4y = 8u(t-2), \qquad y(0) = 0, \quad y'(0) = 0.$



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The Principle of Superposition

Consider the separable differential equation $\frac{dy}{dx} = ky$ where k is a positive constant. A solution curve passes through the points $(0, 5)$ and $(2, 20)$. Determine the value of k and then find the value of y when $x = 3$.

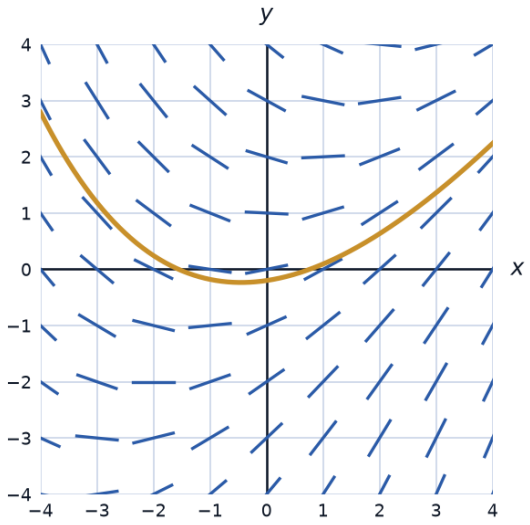
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9 **[Intermediate]** • **Target: General Solution Structure**

Consider the differential equation $\frac{dy}{dx} = \frac{x^2\cos(y)}{1-x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. If there exists a unique solution in a neighborhood of $x = 0$ and $y = \frac{\pi}{2}$, find the value of $y'(0)$.

10 **[Intermediate]** • **Target: Exponential and Sinusoidal Forcing Functions**

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. You are asked to approximate $y(0.2)$ using Euler's method. If the step size $h = 0.1$ is used, and it is known that the exact solution at $x = 0.2$ is $y(0.2) = 1.24$, determine the value of k such that $y(0.2)_{\text{Euler}} = 1.24 - k$.



11 **[Intermediate]** • **Target: The Modification Rule (Resonance / Overlap)**

Find a particular solution of $y'' - 10y' + 25y = e^{5x}x\ln x$ (for $x > 0$) using variation of parameters.

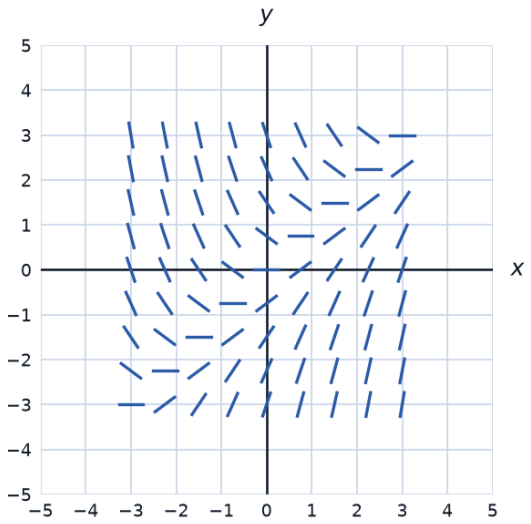
12 **[Intermediate]** • **Target: The Principle of Superposition**

Find the coefficient a_5 of the power series solution $y = \sum a_n x^n$ of $y'' - xy' - y = 0$ with $y(0) = 1$, $y'(0) = 2$.

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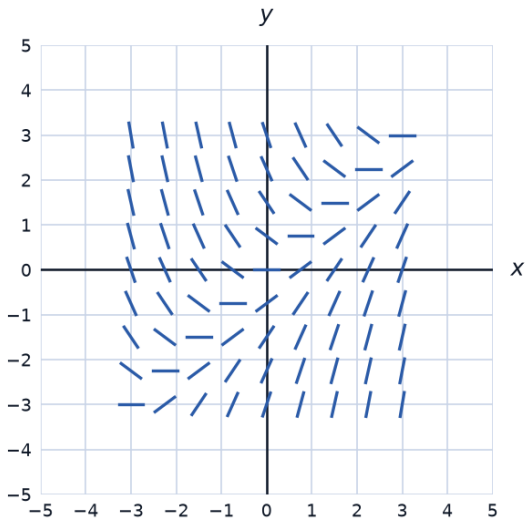
13 **[Intermediate]** • **Target: General Solution Structure**

A harvested population evolves according to $\frac{dP}{dt} = P\left(1 - \frac{P}{10}\right) - 2.1$. Determine the two equilibrium populations, identify which is the extinction threshold, and determine whether a starting population $P(0) = 2$ survives or collapses.



14 **[Intermediate]** • **Target: Exponential and Sinusoidal Forcing Functions**

Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = 1$. Determine the exact time $t > 0$ at which $y(t)$ attains its absolute maximum value.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the stiff decay equation $\frac{dy}{dt} = -100y$ with $y(0) = 1$. Determine the stability threshold h_{crit} for Forward Euler, and compute the value of $y(0.1)$ using Backward (Implicit) Euler with step size $h = 0.05$.

16 [Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the differential equation $\frac{dy}{dx} = \frac{\cos(x)}{y^2}$ for $y > 0$ and $x \in [0, \frac{\pi}{2}]$. If $y(0) = 2$, find the value of $y(\frac{\pi}{2})$.

17 [Challenging] • Target: General Solution Structure • Cross-Lesson: 5.01 Variation of Parameters

Use the Laplace transform and the Convolution Theorem to solve the integral equation $y(t) = t^2 + \int_0^t e^{t-\tau} y(\tau)\, d\tau$.

18 [Challenging] • Target: Exponential and Sinusoidal Forcing Functions • Cross-Lesson: 1.01 Separable Equations

Consider the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Use the Euler method with step size $h = 0.5$ to approximate $y(1)$.

19 [Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $\frac{dy}{dx} = \frac{1+y^2}{x^2-4}$ on $(-2, 2)$ with $y(0) = 1$. At what value of x does $y(x) \to +\infty$?

20 [Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $x^2y' + xy = \frac{1}{x^2+1}$ for $x > 0$, with $y(1) = \frac{\pi}{4}$. What is $y(\sqrt{3})$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: General Solution Structure

What is the structure of the general solution to a linear nonhomogeneous equation $L[y] = g(x)$?

a. $y(x) = y_c(x) + y_p(x)$, where y_c is the complementary solution and y_p is any particular solution

b. $y(x) = y_c(x) \cdot y_p(x)$

c. $y(x) = y_p(x)$ only

d. $y(x) = c_1 y_p(x) + c_2 y_c(x)$

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22 [Foundational] • Target: General Solution Structure

Find a particular solution y_p to $y'' - 5y' + 6y = 12$.

a. $y_p = 2$

b. $y_p = 12$

c. $y_p = 2x$

d. $y_p = 6$

23 [Foundational] • Target: General Solution Structure

What is the trial particular solution y_p for $y'' + 4y = 3x^2 - 1$?

a. $y_p = Ax^2 + Bx + C$

b. $y_p = Ax^2$

c. $y_p = Ax^2 + B$

d. $y_p = Ax^3 + Bx^2$

24 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

Find a particular solution to $y'' + y = 4e^{2x}$.

a. $y_p = \frac{4}{5}e^{2x}$

b. $y_p = 4e^{2x}$

c. $y_p = 2e^{2x}$

d. $y_p = \frac{1}{2}e^{2x}$

25 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

What is the correct undetermined coefficients form for $y'' + 9y = 5\cos(2x)$?

a. $y_p = A\cos(2x) + B\sin(2x)$

b. $y_p = A\cos(2x)$

c. $y_p = Ax\cos(2x)$

d. $y_p = A\sin(2x)$

26 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

Find y_p for $y'' + 9y = 5\cos(2x)$.

a. $y_p = \cos(2x)$

b. $y_p = 5\cos(2x)$

c. $y_p = \frac{5}{13}\cos(2x)$

d. $y_p = \sin(2x)$

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27 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

When must a trial particular solution y_p be multiplied by x (or x^2)?

a. When any term in the trial solution is already a solution of the associated homogeneous equation

b. Whenever the forcing function is a polynomial

c. When the coefficient of y' is negative

d. Only when the right-hand side is zero

28 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

For $y'' - 4y' + 4y = 3e^{2x}$, the homogeneous solution is $y_c = (c_1 + c_2x)e^{2x}$. What is the correct trial form for y_p?

a. $y_p = Ax^2e^{2x}$

b. $y_p = Ae^{2x}$

c. $y_p = Axe^{2x}$

d. $y_p = Ax^3e^{2x}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Modification Rule (Resonance / Overlap)

What physical phenomenon occurs when an undamped mechanical system is driven at its natural resonance frequency ($y'' + \omega_0^2 y = F_0\cos(\omega_0 t)$)?

a. Resonance, where amplitude grows linearly with time ($y_p = Ct\sin(\omega_0 t)$)

b. Immediate decay to zero

c. Critical damping

d. Chaotic motion

30 [Intermediate] • Target: The Principle of Superposition

If y_{p1} solves $L[y] = g_1(x)$ and y_{p2} solves $L[y] = g_2(x)$, what solves $L[y] = 2g_1(x) - 3g_2(x)$?

a. $2y_{p1} - 3y_{p2}$

b. $y_{p1}y_{p2}$

c. $\frac{y_{p1}}{y_{p2}}$

d. $y_{p1} + y_{p2}$

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31 [Intermediate] • Target: The Principle of Superposition

Can the Method of Undetermined Coefficients be used to solve $y'' + y = \tan x$?

a. No, because derivatives of $\tan x$ produce infinitely many non-terminating forms; Variation of Parameters is required

b. Yes, with trial solution $A\tan x + B\sec x$

c. Yes, with trial solution $A\sin x/\cos x$

d. Only if x is small

32 [Intermediate] • Target: The Principle of Superposition

Find a particular solution to $y'' + 4y = 8x + 12$.

a. $y_p = 2x + 3$

b. $y_p = 2x$

c. $y_p = 4x + 3$

d. $y_p = 2x^2 + 3x$

33 [Intermediate] • Target: General Solution Structure

Consider the second-order nonhomogeneous linear differential equation $y'' - 2y' + y = e^x\sin(x)$. Let $y_p(x)$ be a particular solution obtained using the method of undetermined coefficients. Determine the value of $y_p''(\pi) - 2y_p'(\pi) + y_p(\pi)$.

a. 0

b. e^{π}

c. $-\pi e^{\pi}$

d. $e^{\pi}\sin(\pi)$

34 [Intermediate] • Target: Exponential and Sinusoidal Forcing Functions

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

a. 2

b. $2x$

c. $2\sin(2x)$

d. $2\cos(2x)$

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35 [Intermediate] • Target: The Modification Rule (Resonance / Overlap)

Consider the nonhomogeneous linear second-order differential equation $y'' + y = 5$. If $y_p(x) = A$ is a particular solution, what is the value of A?

a. 1

b. 5

c. 1/5

d. 0

36 [Intermediate] • Target: The Principle of Superposition

Consider the second-order linear nonhomogeneous differential equation: $y''(x) - 3y'(x) + 2y(x) = 4$. Find the particular solution, $y_p(x)$, using the method of Undetermined Coefficients.

a. $y_p(x) = 2$

b. $y_p(x) = 2x$

c. $y_p(x) = 4$

d. $y_p(x) = 4x$

37 [Intermediate] • Target: General Solution Structure

An undamped oscillator satisfies $y'' + 4y = 3\cos 2t$ (forcing at exactly the natural frequency), $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/4)$.

a. $-3\pi/16$

b. $3\pi/16$

c. $3\pi/8$

d. $\pi/4$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Exponential and Sinusoidal Forcing Functions • Cross-Lesson: 1.01 Separable Equations

Find a particular solution of $y'' - y' - 6y = e^{2x} + 18$.

a. $14e^2x - 3$

b. $-14e^2x - 3$

c. $-14xe^2x - 3$

d. $-14e^2x + 3$

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39 [Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Find a particular solution of $y'' - 4y' + 3y = 6x + 4$.

a. $2x - 4$

b. $2x + 2$

c. $2x + 4$

d. $4x + 4$

40 [Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' - y' - 2y = 3e^{2x}$, $y(0) = 1$, $y'(0) = 6$. Find $y(1)$.

a. $3e^2 + e^{-1}$

b. $e^2 - e^{-1}$

c. $3e^2 - e$

d. $3e^2 - e^{-1}$

41 [Challenging] • Target: General Solution Structure • Cross-Lesson: 5.01 Variation of Parameters

Solve $y'' + 9y = 2\sin 3t$, $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/3)$.

a. $\pi/9$

b. $-\pi/9$

c. $\pi/3$

d. $2\pi/9$

42 [Challenging] • Target: Exponential and Sinusoidal Forcing Functions • Cross-Lesson: 1.01 Separable Equations

Solve $y'' - 7y' + 12y = -56e^{-4x} - 144x$, $y(0) = -10$, $y'(0) = -15$. Find $y(1)$.

a. $e^{-4} + 19 + e^3 + e^4$

b. $-2e^4 - 2e^3 - 38 - \frac{2}{e^4}$

c. $-\frac{e^4}{2} - \frac{e^3}{2} - \frac{19}{2} - \frac{1}{2e^4}$

d. $-e^4 - e^3 - 19 - e^{-4}$

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43 [Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' + 3y' + 2y = 20e^{3x} - 4x$, $y(0) = -1$, $y'(0) = 8$. Find $y(1)$.

a. $-e^3 - 1 + \frac{2}{e^2} + \frac{3}{e}$

b. $-\frac{3}{e} - \frac{2}{e^2} + 1 + e^3$

c. $-\frac{6}{e} - \frac{4}{e^2} + 2 + 2e^3$

d. $-\frac{3}{2e} - e^{-2} + \frac{1}{2} + \frac{e^3}{2}$

44 [Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' + 4y' + 3y = -35e^{4x} - 9x$, $y(0) = 4$, $y'(0) = -14$. Find $y(1)$.

a. $-e^4 - \frac{2}{e} + \frac{3}{e^3} + 1$

b. $-1 - \frac{3}{e^3} + \frac{2}{e} + e^4$

c. $-2e^4 - \frac{4}{e} + \frac{6}{e^3} + 2$

d. $-\frac{e^4}{2} - e^{-1} + \frac{3}{2e^3} + \frac{1}{2}$

45 [Challenging] • Target: General Solution Structure • Cross-Lesson: 5.01 Variation of Parameters

Solve $y'' - y' - 2y = 10e^{-3x} + 4x$, $y(0) = 3$, $y'(0) = -12$. Find $y(1)$.

a. $-2e^2 - 1 + e^{-3} + \frac{3}{e}$

b. $-\frac{3}{e} - e^{-3} + 1 + 2e^2$

c. $-4e^2 - 2 + \frac{2}{e^3} + \frac{6}{e}$

d. $-e^2 - \frac{1}{2} + \frac{1}{2e^3} + \frac{3}{2e}$

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